

## Problem Set for 6th March 2025

**Exercise 1** Here is the composition rules for linear maps  $U \xrightarrow{f} V \xrightarrow{g} W$ . Show that

1. if  $f$  and  $g$  are linear injections, then so is  $g \circ f$ ;
2. if  $g \circ f$  is a linear injection, then so is  $f$ ;
3. if  $f$  and  $g$  are linear surjections, then so is  $g \circ f$ ;
4. if  $g \circ f$  is a linear surjection, then so is  $g$ .

**Exercise 2** Assume you can prove the following isomorphisms yourself:

$$\Phi : \text{Hom}_{\mathbb{F}}(U_1, V) \times \text{Hom}_{\mathbb{F}}(U_2, V) \rightarrow \text{Hom}_{\mathbb{F}}(U_1 \times U_2, V)$$

$$(f_1, f_2) \mapsto \begin{bmatrix} U_1 \times U_2 & \rightarrow & V \\ (u_1, u_2) & \mapsto & f_1(u_1) + f_2(u_2) \end{bmatrix},$$

and

$$\Psi : \text{Hom}_{\mathbb{F}}(U, V_1) \times \text{Hom}_{\mathbb{F}}(U, V_2) \rightarrow \text{Hom}_{\mathbb{F}}(U, V_1 \times V_2)$$

$$(g_1, g_2) \mapsto \begin{bmatrix} U & \rightarrow & V_1 \times V_2 \\ u & \mapsto & (g_1(u), g_2(u)) \end{bmatrix}.$$

Show that

1.  $\Phi(f_1, f_2)$  is an injection, only if  $(\Rightarrow)$   $f_1$  **and**  $f_2$  are injective;
2.  $\Phi(f_1, f_2)$  is a surjection, if  $(\Leftarrow)$   $f_1$  **or**  $f_2$  is surjective;
3.  $\Psi(g_1, g_2)$  is an injection, if  $(\Leftarrow)$   $g_1$  **or**  $g_2$  is injective;
4.  $\Psi(g_1, g_2)$  is a surjection, only if  $(\Rightarrow)$   $f_1$  **and**  $f_2$  are surjective;
5. **(Optional)** Find counterexamples for converse statements.

**Exercise 3 (optional)** We know that the composition map

$$B : \text{Hom}_{\mathbb{F}}(V, W) \times \text{Hom}_{\mathbb{F}}(U, V) \rightarrow \text{Hom}_{\mathbb{F}}(U, W)$$

$$(g, f) \mapsto B(g, f) := g \circ f$$

is  $\mathbb{F}$ -bilinear. Show that

1. if  $g$  is injective, then  $B(g, -)$  is an injection;
2. if  $f$  is surjective, then  $B(-, f)$  is **still** an injection;
3. if  $g$  is surjective and  $V$  is coflasque, then  $B(g, -)$  is surjective;
4. if  $f$  is injective and  $V$  is flasque, then  $B(-, f)$  is **still** surjective.

## Definition

- Say  $Y$  is flasque, if any linear map on its subspace  $Y^{\text{sub}} \rightarrow X$  extends to  $Y \rightarrow X$ .
- Say  $Y$  is coflasque, if any linear map on its quotient space  $X \rightarrow Y^{\text{quot}}$  extends to  $X \rightarrow Y$ .

Finite dimensional vector spaces are both flasque and co-flasque.

**Exercise 4 (optional)** Equivalent definition of being injective or surjection.

1.  $f$  is injective, if and only if

$$[f \circ g = f \circ h] \iff [g = h].$$

2.  $f$  is surjective, if and only if

$$[g \circ f = h \circ f] \iff [g = h].$$

For **finite dimensional** cases

1.  $f : U \rightarrow V$  is injective, if and only if there is an  $g : V \rightarrow U$  such that

$$\left[ U \xrightarrow{f} V \xrightarrow{g} U \right] = \left[ U \xrightarrow{\text{id}_U} U \right].$$

2.  $f : U \rightarrow V$  is surjective, if and only if there is an  $h : V \rightarrow U$  such that

$$\left[ V \xrightarrow{h} U \xrightarrow{f} V \right] = \left[ V \xrightarrow{\text{id}_V} V \right].$$